

Lecture-12

“VB. Net Lab”

Presented By:

Ms. Jyoti Singh

B.C.A., M.C.A.,P.hd(Pursuing)

BCA-605P

Advanced Technology (Dot Net) Lab

1. Familiarization with IDE.
2. Programming Console applications using VB.NET covering all the aspects

of VB.NET fundamentals

3. Object oriented programming using VB.NET covering objects, Inheritance, Polymorphism,

Constructors, Static Classes, and Interfaces.

4. Programme to illustrate Exception Handling concepts.

5. Programme to illustrate use of Collections.

6. Programme to perform File I/O Operations.

7. Programming Windows applications using VB.NET covering all major controls and components,

Menus, MDI, Event Handling.

8. Creating windows installer.

9. Programme to interact with Database from a Windows Desktop Application.

10. Programming to build web applications using web controls, maintaining state.

Note: The Instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner.

VB.NET Arrays

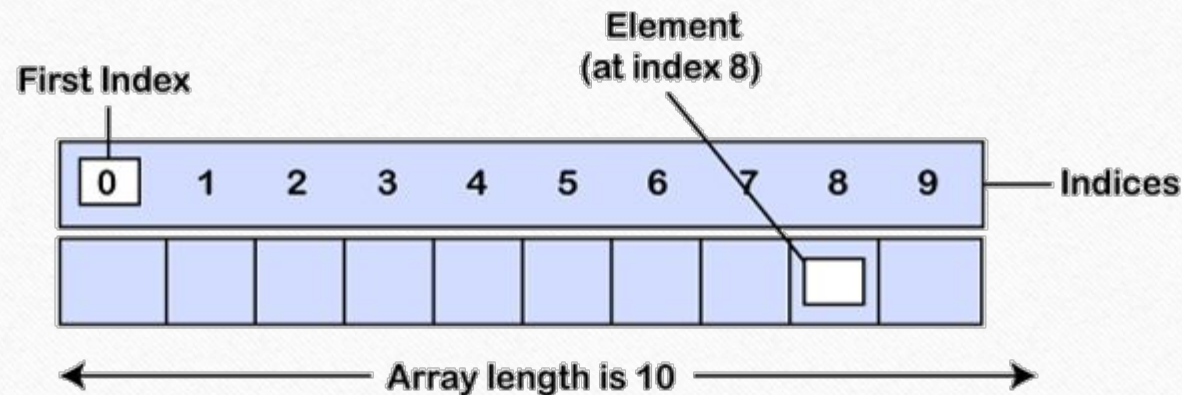
- **Why use Array?**

Array

- An **array** is a linear data structure that is a collection of data elements of the same type stored on a **contiguous memory** location.
- Each data item is called an element of the array.
- It is a fixed size of sequentially arranged elements in computer memory with the first element being at **index 0** and the last element at index **$n - 1$** , where **n** represents the total number of elements in the array.

-
- []-Index, subscript, square bracket, dimension
 - `Int a[5];`
 - 01234

- The following is an illustrated representation of similar data type elements defined in the VB.NET array data structure.



In the above diagram, we store the Integer type data elements in an array starting at index 0.

It will continue to store data elements up to a defined number of elements.

Declaration of VB.NET Array

- We can declare an array by specifying the data of the elements followed by parentheses () in the VB.NET.
- **Dim array_name As [Data_Type] ()**
- In the above declaration, **array_name** is the name of an array, and the **Data_Type** represents the type of element (Integer, char, String, Decimal) that will store contiguous data elements in the VB.NET array.

- Now, let us see the example to declare an array.

- **'Store only Integer values**

- Dim num As Integer() or Dim num(5) As Integer

- **'Store only String values**

- Dim name As String() or Dim name(5) As String

- **' Store only Double values**

- Dim marks As Double()

Initialization of VB.NET Array

- In VB.NET, we can initialize an array with **New** keyword at the time of declaration.
 - For example,
-

'Declaration and Initialization of an array elements with size 6

```
Dim num As Integer() = New Integer(5) { }
```

```
Dim num As Integer() = New Integer(5) {1, 2, 3, 4, 5, 6}
```

Initialize an array with 5 elements that indicates the size of an array

```
Dim arr_name As Integer() = New Integer() {5, 10, 5, 20, 15}
```

Declare an array

```
Dim array1 As Char()
```

```
array1 = New Char() {'A', 'B', 'C', 'D', 'E'}
```

Furthermore, we can also initialize and declare an array using the following ways, as shown below.

```
Dim intData() As Integer = {1, 2, 3, 4, 5}
```

```
Dim intData(5) As Integer
```

```
Dim array_name() As String = {"Peter", "John", "Brock", "James", "Maria"}
```

```
Dim misc() as Object = {"Hello friends", 16c, 12ui, "A"c}
```

```
Dim Emp(0 to 2) As String
```

```
Emp{0} = "Mathew"
```

```
Emp(1) = " Anthony"
```

```
Emp(2) = "Prince"
```

Create a program to add the elements of an array in VB.NET programming language.

Imports System

Module num_Array

Sub Main()

Dim i As Integer, Sum As Integer = 0

'In VB.NET the size of an array is n+1

'Declaration and Initialization of marks() array

Dim marks() As Integer = {58, 68, 95, 50, 23, 89}

Console.WriteLine(" Marks in 6 Subjects")

For i = 0 To marks.Length - 1

Console.WriteLine(" Marks {0}", marks(i))

Sum = Sum + marks(i)

Next

Console.WriteLine(" Grand total is {0}", Sum)

Console.WriteLine(" Press any key to exit...")

Console.ReadKey()

End Sub

End Module

Create a program to take input values from the user and display them in VB.NET programming language.

```
Imports System
```

```
Module Input_array
```

```
Sub Main()
```

```
    'Definition of array
```

```
    Dim arr As Integer() = New Integer(5) {}
```

```
    For i As Integer = 0 To 5
```

```
        Console.WriteLine(" Enter the value for arr[{0}] : ", i)
```

```
        arr(i) = Console.ReadLine() ' Accept the number in array
```

```
    Next
```

```
    Console.WriteLine(" The array elements are : ")
```

```
    ' Definition of For loop
```

```
    For j As Integer = 0 To 5
```

```
        Console.WriteLine("{0}", arr(j))
```

```
    Next
```

```
    Console.WriteLine(" Press any key to exit...")
```

```
    Console.ReadKey()
```

```
End Sub
```

```
End Module
```

Multidimensional Array

- In VB.NET, a multidimensional array is useful for storing more than one dimension in a tabular form, such as rows and columns.
- The multidimensional array support two or three dimensional in VB.NET.

Declaration of Multidimensional Array

Declaration of two-dimensional array

```
Dim twoDimenArray As Integer( , ) = New Integer(3, 2) { }
```

Or Dim arr(5, 3) As Integer

Representation of Three Dimensional array

```
Dim arrThree(2, 4, 3) As Integer
```

```
Or Dim arr1 As Integer( , , ) = New Integer(5, 5, 5) { }
```

-
- In the above representation of multidimensional, we have created
2-dimensional array **twoDimenArray** with 3 rows and 2 columns
and 3-dimensional array with three dimensions 2, 4, and 3.

Initialization of Multidimensional Array

' Initialization of Two Dimensional Array

```
Dim intArray As Integer( , ) = New Integer( 3, 2) { {4, 5}, {2, 3}, {6, 7} }
```

```
Dim intArray( , ) As Integer = { {5, 4}, {3, 2}, {4, 7} }
```

' Initialization of Three Dimensional Array

```
Dim threeDimen(3, 3, 2 ) As Integer = { {{1, 3, 2}, {2, 3, 4}}, {{5, 3, 6}, {3, 4, 5}}, {{1, 2, 2}, {5, 2, 3} }}
```

Multidimensional Array Example

● Program

Thank You